

A short note on polynomial interpolation

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April 28, 2026

1 Introduction

Interpolation is the methodology of constructing new points between known data points. We often confront situations where only a limited amount of data is accessible and it is necessary to estimate values between two consecutive data points. In finance, as only a finite set of instruments are traded in financial markets, it plays a key role to construct a sensible curve or surface from discrete observable quantities.

2 Splines

We start by formally defining a spline function of degree $k - 1 \geq 0$.

Definition 2.1. A spline function $S(x)$ of degree $k - 1 \geq 0$ on a grid:

$$\Delta = \{a = x_0 < x_1 < \dots < x_n = b\}$$

of distinct knots is a real function with the following properties:

1. For $x \in [x_i, x_{i+1}]$, $S(x)$ is a polynomial of degree $< k$.
2. Its first $k - 2$ derivatives are continuous, that is $S(x) \in C^{k-2}[a, b]$.

The space of all spline functions of degree $k - 1 \geq 0$ is denoted by $S_{\Delta, k}$. From the definition, it follows that if $S_1(x)$ and $S_2(x)$ are spline functions, so is $c_1 S_1(x) + c_2 S_2(x)$. Clearly, $S_{\Delta, k}$ is a linear vector space.

3 Cubic Spline Interpolant

A cubic spline uses cubic polynomials of degree 3 to interpolate between successive knots. A cubic spline interpolant $S(x)$ satisfies the following conditions:

1. Each piece or segment $S_j(x)$ on the interval $[x_j, x_{j+1}]$ is a cubic polynomial.
2. $S_j(x_j) = y_j$ for each $j = 0, 1, \dots, n$.
3. $S_j(x_{j+1}) = S_{j+1}(x_{j+1})$, $S'_j(x_{j+1}) = S'_{j+1}(x_{j+1})$ and $S''_j(x_{j+1}) = S''_{j+1}(x_{j+1})$.

Condition (3) is just a translation of the requirement that the whole spline should be $k - 2$ times differentiable and these derivatives must be continuous. As a crude approximation, we know that a continuous curve is one that can be drawn without lifting pen from paper. The individual segments of $S(x)$ are cubics, so they are continuous themselves. The only consideration is the continuity at knot points.

For a curve to be continuous at any given knot-point c , whether we approach it from the left or we approach it from the right, the spline function $S(x)$ must approach $S(c)$ regardless. Suppose the left piece is S_j and the right piece is S_{j+1} . This motivates $S_j(x_{j+1}) = S_{j+1}(x_{j+1})$. A similar argument follows for the first and second derivatives because we would like those to be continuous.

4 Construction of the Cubic Spline

To construct the cubic spline interpolant on a grid Δ divided into n subintervals, we require n segments $S_0, S_1, \dots, S_{n-1}$. In line, with condition (1), the segment S_j is a cubic polynomial of the form:

$$S_j(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 + d_j(x - x_j)^3 \quad (1)$$

and has 4 constants. Therefore, we have to determine the values of $4n$ constants. The first and second derivatives of $S_j(x)$ are given by:

$$S'_j(x) = b_j + 2c_j(x - x_j) + 3d_j(x - x_j)^2 \quad S''_j(x) = 2c_j + 6d_j(x - x_j) \quad (2)$$

So, $S_{j+1}(x_{j+1}) = a_{j+1}$, $S'_{j+1}(x_{j+1}) = b_{j+1}$ and $S''_{j+1}(x_{j+1}) = 2c_{j+1}$.

Also, the quantity $(x_{j+1} - x_j)$ appears so often, we let $h_j = (x_{j+1} - x_j)$. Applying conditions (2) and (3), we obtain:

$$S_{j+1}(x_{j+1}) = a_{j+1} = y_{j+1} = S_j(x_{j+1}) = a_j + b_j h_j + c_j h_j^2 + d_j h_j^3 \quad (3)$$

$$S'_{j+1}(x_{j+1}) = b_{j+1} = S'_j(x_{j+1}) = b_j + 2c_j h_j + 3d_j h_j^2 \quad (4)$$

$$(1/2)S''_{j+1}(x_{j+1}) = c_{j+1} = (1/2)S''_j(x_{j+1}) = c_j + 3d_j h_j \quad (5)$$

for each $j = 0, 1, \dots, n-1$.

Solving for d_j in equation (5), and substituting this value in equations (3) and (4) gives, for each $j = 0, 1, \dots, n-2$, the new equations:

$$\begin{aligned} a_{j+1} &= a_j + b_j h_j + c_j h_j^2 + \left(\frac{c_{j+1} - c_j}{3h_j} \right) h_j^3 \\ &= a_j + b_j h_j + c_j h_j^2 - c_j \frac{h_j^2}{3} + \frac{h_j^2}{3} c_{j+1} \\ &= a_j + b_j h_j + c_j h_j^2 + \frac{h_j^2}{3} (2c_j + c_{j+1}) \end{aligned} \quad (6)$$

$$\begin{aligned} b_{j+1} &= b_j + 2c_j h_j + 3 \left(\frac{c_{j+1} - c_j}{3h_j} \right) h_j^2 \\ &= b_j + 2c_j h_j + (c_{j+1} - c_j) h_j \\ &= b_j + (c_j + c_{j+1}) h_j \end{aligned} \quad (7)$$

The final relationship is obtained by solving equation (6) first for b_j :

$$b_j = \frac{1}{h_j} (a_{j+1} - a_j) - \frac{h_j}{3} (2c_j + c_{j+1}) \quad (8)$$

and then with a reduction of the index, for b_{j-1} gives:

$$b_{j-1} = \frac{1}{h_{j-1}} (a_j - a_{j-1}) - \frac{h_{j-1}}{3} (2c_j + c_{j+1}) \quad (9)$$

Substituting these values into the equation derived from (7), with the index reduced by one gives the linear system of equations:

$$\begin{aligned}
b_j &= b_{j-1} + (c_{j-1} + c_j)h_{j-1} \\
\frac{1}{h_j}(a_{j+1} - a_j) - \frac{h_j}{3}(2c_j + c_{j+1}) &= \frac{1}{h_{j-1}}(a_j - a_{j-1}) - \frac{h_{j-1}}{3}(2c_{j-1} + c_j) + (c_{j-1} + c_j)h_{j-1} \\
\frac{1}{h_j}(a_{j+1} - a_j) - \frac{1}{h_{j-1}}(a_j - a_{j-1}) &= \frac{h_j}{3}(2c_j + c_{j+1}) - \frac{h_{j-1}}{3}(2c_{j-1} + c_j) + (c_{j-1} + c_j)h_{j-1} \\
\frac{3}{h_j}(a_{j+1} - a_j) - \frac{3}{h_{j-1}}(a_j - a_{j-1}) &= 2c_j h_j + c_{j+1} h_j - 2c_{j-1} h_{j-1} - c_j h_{j-1} + 3c_{j-1} h_{j-1} + 3c_j h_{j-1} \\
\frac{3}{h_j}(a_{j+1} - a_j) - \frac{3}{h_{j-1}}(a_j - a_{j-1}) &= 2c_j h_j + c_{j+1} h_j + c_{j-1} h_{j-1} + 2c_j h_{j-1} \\
\frac{3}{h_j}(a_{j+1} - a_j) - \frac{3}{h_{j-1}}(a_j - a_{j-1}) &= c_{j+1} h_j + 2h_j(c_j + c_{j-1}) + c_{j-1} h_{j-1}
\end{aligned} \tag{10}$$

for each $j = 1, 2, \dots, n-1$. This system involves only the $\{c_j\}_{j=0}^n$ unknowns. The values of $\{h_j\}_{j=0}^n$ and $\{a_j\}_{j=0}^n$ are given respectively by the spacing of the knots $\{x_j\}_{j=0}^n$ and the values $\{y_j\}_{j=0}^n$ at the knot-points. Once the values of $\{c_j\}_{j=0}^n$ are determined, it is a simple task to find the remainder of the constants $\{b_j\}_{j=0}^n$ and $\{d_j\}_{j=0}^b$ from the equations (5) and (7).